

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K₄ Fe (CN)₆. After filtering, the precipitate is dried and weighed. The reactions which occur are $\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$ $2\text{ZnCl}_2 + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Zn}_2\text{Fe}(\text{CN})_6 + 4\text{KCl}$ If a 2 gram sample of rock yields 0.969 gram of Zn₂Fe(CN)₆, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn₂ Fe (CN)₆ = 342.6.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<think>

- $\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$: Zn is stoichiometrically present in ZnS and ZnCl₂.

- $\text{ZnCl}_2 + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Zn}_2\text{Fe}(\text{CN})_6 + 4\text{KCl}$: Zn is stoichiometrically present in Zn₂Fe(CN)₆.

- Therefore the amount of Zn₂Fe(CN)₆ produced is stoichiometrically proportional to Zn in ZnS.

- Molecular weight of Zn₂Fe(CN)₆ = 2 * Zn + Fe + 6 * C + 6 * N = 2 * 65.4 + 52 + 6 * 12 + 6 * 14 = 2 * 65.4 + 66 + 84 = 270.8 g which is Zn part.

- Zn part in Zn₂Fe(CN)₆ : in Zn₂Fe(CN)₆ = 2 * 65.4 / 342.6 * 100 ~ = 37.8 %.

- Atomic weight Zn / Molecular weight Zn₂Fe(CN)₆ * 100 ~ = 37.8 / 100 * 100 ~ 10.5 %.

</think>

<answer>

A. 10 % Zn

</answer>